Solution

d = 0.15 mm = 0.15× 10^{-3} m; D = 2 m; λ = 450 nm = 450 × 10^{-9} m; RI = 4/3

(a) Equation for n^{th} bright fringe,

$$y_n = n\frac{\lambda D}{d}$$

Distance of 2nd bright fringe,

$$y_2 = 2\times\frac{450\times10^{-9}\times2}{0.15\times10^{-3}}$$

$$y_2 = 12\times10^{-3}\,\text{m} = 12\ \text{mm}$$

Equation for n^{th} dark fringe,

$$y_n = \frac{(2n-1)}{2}\frac{\lambda D}{d}$$

Distance of 3rd dark fringe,

$$y_3 = \frac{5}{2}\times\frac{450\times10^{-9}\times2}{0.15\times10^{-3}}$$

$$y_2 = 15\times10^{-3}\,\text{m} = 15\ \text{mm}$$

(b) Equation for fringe width, $\beta = \frac{\lambda D}{d}$

Substituting, $\beta = \frac{450\times10^{-9}\times2}{0.15\times10^{-3}}$

$$\beta = 6\times10^{-3}\,\text{m} = 6\ \text{mm}$$

(c) The fringe width will increase as D is increased, $\beta = \frac{\lambda D}{d}$ (or) $\beta \propto D$

(d) The fringe width will decrease as the setup is immersed in water of refractive index 4/3

$$\beta = \frac{\lambda D}{d} \quad \text{(or)} \quad \beta \propto \lambda$$

The wavelength will decrease in a medium. Hence, $\beta \propto \lambda$ and $\beta' \propto \lambda'$

We know that, $\lambda' = \frac{\lambda}{RI}$

$$\frac{\beta'}{\beta} = \frac{\lambda'}{\lambda} = \frac{\lambda/RI}{\lambda} = \frac{1}{RI} \quad \text{(or)} \quad \beta' = \frac{\beta}{RI} = \frac{6\times10^{-3}}{4/3}$$

$$\beta' = 4.5\times10^{-3}\,\text{m} = 4.5\ \text{mm}$$

7.3.5 Interference in white light (polychromatic light)

When a white light (polychromatic light) is used in interference experiment, coloured fringes of varied thickness will be formed on the screen. This is because, different colours have different wavelengths. However, the central fringe (or) 0th fringe will always be bright and white in colour, because all the colours falling at the point O will have no path difference with each other. Hence, only constructive interference is possible at O for all the colours.

EXAMPLE 7.7

Lights of two wavelengths 560 nm and 420 nm are used in Young's double slit experiment. Find the least distance from the central fringe where the bright fringes of the two wavelengths coincide. Given D = 1 m and d = 3 mm.

Solution

$$\lambda_1 = 560\ \text{nm} = 560\times10^{-9}\,\text{m}$$

$$\lambda_2 = 420\ \text{nm} = 420\times10^{-9}\,\text{m}$$

$$D = 1\,\text{m};\ d = 3\ \text{mm} = 3\times10^{-3}\ \text{m}$$

Here, n and λ are inversely proportional for a given y.

Here, n^{th} order bright fringe of longer wavelength λ_1 coincides with $(n+1)^{th}$ order bright fringe of shorter wavelength λ_2.

Equation for n^{th} bright fringe is, $y_n = n\frac{\lambda D}{d}$

Here, $n\frac{\lambda_1 D}{d} = (n+1)\frac{\lambda_2 D}{d}$ (as $\lambda_1 > \lambda_2$)

$$n\lambda_1 = (n+1)\lambda_2 \,(\text{or})\, \frac{\lambda_1}{\lambda_2} = \frac{(n+1)}{n};\ 1+\frac{1}{n} = \frac{\lambda_1}{\lambda_2}$$

$$1+\frac{1}{n} = \frac{560\times10^{-9}}{420\times10^{-9}} \quad \text{(or)} \quad 1+\frac{1}{n} = \frac{4}{3}$$

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